Republic of the Philippines
BATANGAS STATE UNIVERSITY
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College of Engineering – Department of Electrical Engineering

CURRICULUM

Bachelor of Science in Computer Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 87 s. 2017, CMO No. 4 s. 2018 and CMO No. 20, s. 2013, CMO 39, s. 2021, CMO 40, s. 2021

Curriculum Description

The Bachelor of Science in Computer Engineering (BSCpE) is a program that embodies the Science and technology of design, development, implementation, maintenance and integration of software and hardware components in modern computing systems and computer-controlled equipment.

Graduates of Computer Engineering may work as Software Engineer, Hardware Engineer, Network Engineer, System Administrator, Database Administrator, Cyber Security Specialist, Artificial Intelligence/Machine Learning Specialist, etc.

Program Education Objectives

The Computer engineering alumni three to five years after graduation shall:

PEO 1: Practice as specialist in solving complex computer engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.

PEO 2: Assume leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.

PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex Computer engineering problems;

SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex Computer engineering problems by applying principles of engineering, science, and mathematics;

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SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex Computer engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex Computer engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Differential Equations	3	0	3
Probability and Statistics for AI	3	0	3
Subtotal			**14**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers 1	3	3	4
Physics for Engineers 2	2	3	3
Subtotal			**11**
C. Basic Engineering Sciences			
Computer-Aided Design	0	3	1
Engineering Economics	3	0	3
Technopreneurship	3	0	3
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer Programming	0	3	1
Engineering Management	3	0	2
Subtotal			**12**
D. Allied Courses			
Fundamental of Electrical Circuits	3	3	4

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Electronic Circuits: Devices and Analysis	3	3	4
Subtotal			**8**
E. Professional Courses			
CpE Discipline, Professional Practice and laws	3	0	3
Discrete Mathematics	3	0	3
Programming Logic and Design	0	6	2
Advanced Engineering Mathematics for CpE	3	0	3
Object-Oriented Programming	0	6	2
Numerical Methods	3	0	3
Logic Circuits and Design	3	3	4
Data Structures and Algorithms	0	6	2
Introduction to Networks, Data and Digital Communications	2	3	3
Fundamentals of Mixed Signals and Sensors	3	0	3
Feedback and Control Systems	3		3
Microprocessors	3	3	4
Software Design	3	3	4
Digital Signal Processing	3	3	4
Computer Architecture and Organization	3	3	4
Connecting Networks and Security	3	3	4
Introduction to HDL	0	3	1
Emerging Technologies in CpE	3	0	3
Computer Engineering Drafting and Design	0	3	1
Operating Systems	3	0	3
Embedded Systems	3	3	4
Seminars and Fieldtrips	0	6	2
CpE Practice and Design 1	0	3	1
CpE Practice and Design 2	0	6	2
Comprehensive Examination for CpE	0	6	2
Methods of Research	3	0	3
Basic Occupational Safety and Health	3	0	3
OJT/Internship (320 hrs)	**320 hours**		4
Subtotal			**80**
F. Elective Courses (Choose any 3)			
Track1: Artificial Intelligence			
-Introduction to Artificial Intelligence	2	3	3
-Machine Learning	2	3	3
-Deep Learning	2	3	3
Track2: Big Data Analytics			
-Introduction to Cloud	2	3	3
-Big Data Analytics	2	3	3
-Data Mining	2	3	3
Subtotal			**9**
TOTAL TECHNICAL COURSES			**134**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3

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Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal			**24**
B. General Education Elective and Mandated Courses			
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	2	3	3
The Entrepreneurial Mind	3	0	3
Life and Works of Rizal	3	0	3
Subtotal			**12**
C. Mandated Courses			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal			**14**
TOTAL NON-TECHNICAL COURSES			**50**
GRAND TOTAL			**184**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	14
B. Natural/Physical Sciences	11
C. Basic Engineering Sciences	12
D. Allied Courses	8
E. Professional Courses	80
F. Elective Courses	9
II. NON-TECHNICAL COURSES	
A. General Education Courses	24
B. General Education Electives	12
C. Mandated Courses	14
GRAND TOTAL	**184**

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CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING
Reference CMOs: CMO No. 87 s. 2017, CMO No. 4 s. 2018 and CMO No. 20, s. 2013, CMO 39, s. 2021, CMO 40, s. 2021

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 101	Introduction to Engineering	1	0	3			TC-C
GEd 108	Art Appreciation	3	3	0			NTC-A
GEd 104	The Contemporary World	3	3	0			NTC-A
CpE 101	Computer Programming	1	0	3			TC-C
GEd 105	Readings in Philippine History	3	3	0			NTC-A
MATH 101	Differential Calculus	4	4	0			TC-A
CHEM 101	Chemistry for Engineers	4	3	3			TC-B
PATHFit 1	Movement Competency Training	2	2	0			NTC-C
NSTP 111	National Service Training Program 1	3	3	0			NTC-D
	Total	**24**	**21**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 102	Engineering Drawing	1	0	3			NTC-C
GEd 101	Understanding the Self	3	3	0			NTC-A
GEd 102	Mathematics in the Modern World	3	3	0			NTC-A
GEd 106	Purposive Communication	3	3	0			NTC-A
GEd 109	Science, Technology and Society	3	3	0			NTC-A
MATH 102	Integral Calculus	4	4	0	MATH 101		TC-A
PHYS 102	Physics for Engineers 1	4	3	3	MATH 101	MATH 102	TC-B
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		NTC-C
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC-D
	Total	**26**	**24**	**6**			

FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GEd 103	Life and Works of Rizal	3	3	0			NTC-B
GEd 107	Ethics	3	3	0			NTC-A
PHYS 103	Physics for Engineers 2	3	2	3	PHYS 102; MATH 102		TC-B
	TOTAL	**9**	**8**	**3**			

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SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 201	CpE Discipline, Professional Practice and laws	3	3	0			TC-E
CpE 202	Discrete Mathematics	3	3	0	MATH 101		TC-E
CpE 203	Programming Logic and Design	2	0	6			TC-E
GEdE 101	The Entrepreneurial Mind	3	3	0			NTC-B
EE 212	Fundamentals of Electrical Circuits	4	3	3	PHY 103		TC-D
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102		TC-C
ENGG 104	Engineering Economics	3	3	0	MATH 102		TC-C
AI 101	Linear Algebra for AI	3	3	0			NTC-B
MATH 103	Differential Equations	3	3	0	MATH 102		TC-A
PATHFit 3	Choice of dance, sports, Martial Arts, Group exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**27**	**23**	**12**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 204	Advance Engineering Mathematics for CpE	3	3	0	2nd Year Standing		TC-E
CpE 205	Object-Oriented Programming	2	0	6	CpE 203		TC-E
	Elective Course 1	3	2	3	2nd Year Standing		TC-F
ECE 405	Electronic Circuits: Devices and Analysis	4	3	3	EE 212		TC-D
ENGG 114	Basic Occupational Safety and Health	3	3	0			TC-E
MATH 104	Numerical Methods	3	3	0	Math 103		TC-E
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		TC-A
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**23**	**19**	**12**			

THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 302	Logic Circuits and Design	4	3	3	ECE 405		TC-E
CpE 303	Data Structures and Algorithms	2	0	6	CpE 205		TC-E
CpE 304	Introduction to Networks,	3	2	3	ECE 405	CpE 302	TC-E

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	Data and Digital Communications						
CpE 305	Fundamentals of Mixed Signals and Sensors	3	3	0	ECE 405		TC-E
CpE 306	Feedback and Control Systems	3	3	0	MATH 104; EE 212		TC-E
ENGG 116	Methods of Research	3	3	0	AI 102, 3rd Year Standing		TC-E
	Elective Course 2	3	2	3	Elective Course 1		TC-F
	Total	**21**	**16**	**15**			

THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 308	Microprocessors	4	3	3	CpE 302		TC-E
CpE 309	Software Design	4	3	3	CpE 303		TC-E
CpE 310	Digital Signal Processing	4	3	3	CpE 305; CpE 306		TC-E
CpE 311	Computer Architecture and Organization	4	3	3		CpE 308	TC-E
CpE 312	Connecting Networks and Security	4	3	3	CpE 304		TC-E
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CpE 101		NTC-B
	Total	**23**	**17**	**18**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 313	Introduction to HDL	1	0	3	CpE 203, CpE 302		TC-E
CpE 314	Emerging Technologies in CpE	3	3	0	3rd Year Standing		TC-E
CpE 315	Computer Engineering Drafting and Design	1	0	3	ECE 405		TC-E
	Total	**5**	**3**	**6**			

FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 401	Operating Systems	3	3	0	CpE 303		TC-E
CpE 402	Embedded Systems	4	3	3	CpE 308		TC-E
CpE 403	Seminars and Fieldtrips	2	0	6	4th Year Standing		TC-E
CpE 404	CpE Practice and Design 1	1	0	3	ENGG 116 , CpE 308		TC-E
ENGG 106	Engineering Management	2	2	0	4th Year Standing		TC-C
ENGG 105	Technopreneurship	3	3	0	4th Year Standing		TC-C
	Elective Course 3	3	2	3	Elective Course 2		TC-F
	Total	**18**	**13**	**15**			

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FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 117	OJT/Internship (320 hrs)	4	320 hours		4th Year Standing		TC-E
CpE 405	CpE Practice and Design 2	2	0	6	CpE 404		TC-E
CpE 406	Comprehensive Examination for CpE	2	0	6	Graduating		TC-E
	Total	**8**	**0**	**12**			

TOTAL UNITS: 184

CpE Elective Tracks:

Track 1: Artificial Intelligence

CpEE 401 Introduction to Artificial Intelligence
CpEE 402 Machine Learning
CpEE 403 Deep Learning

Track 2: Big Data Analytics

CpEE 404 Introduction to Cloud
CpEE 405 Big Data Analytics
CpEE 406 Data Mining